Levetiracetam

Mechanism of action

- Levetiracetam is an anticonvulsant. It is thought to inhibit seizure activity by blocking some calcium channels and also binding synaptic proteins to modulate neurotransmitter release.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Seizure that continues or recurs after two doses of parenteral midazolam.
- ✓ Seizure activity associated with severe TBI, even if seizure activity has ceased following midazolam.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

Dosage

- 2 g IV for an adult, this may be repeated once if the seizure continues or recurs ten minutes after the first dose.
- See the paediatric drug dose tables for a child. This may be only administered once.

Administration

- Draw up the required dose.

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- Administer this IV over 5 minutes preferably into a running line.
- Do not administer IM.

Common adverse effects

- None.

Usual onset of effect

- 10 minutes.

Usual duration of effect

- 6-8 hours.

Usual preparation

- Ampoule containing 500 mg in 5 ml.

Pharmacokinetics

- Levetiracetam is excreted by the kidneys.
- There are no significant effects from renal impairment on the initial (loading) dose.

Common interactions

- None.

More Information

Version history

9 June 2025 1.0.5.3
Update: Indications updated.

28 October 2024 1.0.5
Update: Indications updated.

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11 September 2023 1.0.4

New: Guideline released.